create their own picture of the world, and even if this generation fails to synthesise all the data into a coherent and meaningful story of the world, there will be plenty of time to construct a good synthesis in the future. We have now run out of time. The decisions we will take in the next few decades will shape the future of life itself, and we can take these decisions based only on our present world view. If this generation lacks a comprehensive view of the cosmos, the future of life will be decided at random.

The heat is on

Besides information, most schools also focus too much on providing pupils with a set of predetermined skills such as solving differential equations, writing computer code in C++, identifying chemicals in a test tube, or conversing in Chinese. Yet since we have no idea how the world and the job market will look in 2050, we don't really know what particular skills people will need. We might invest a lot of effort teaching kids how to write in C++ or how to speak Chinese, only to discover that by 2050 AI can code software far better than humans, and a new Google Translate app enables you to conduct a conversation in almost flawless Mandarin, Cantonese or Hakka, even though you only know how to say '*Ni hao*.'

So what should we be teaching? Many pedagogical experts argue that schools should switch to teaching 'the four Cs' – critical thinking, communication, collaboration and creativity.[3] More broadly, schools should downplay technical skills and emphasise general-purpose life skills. Most important of all will be the ability to deal with change, to learn new things, and to preserve your mental balance in unfamiliar situations. In order to keep up with the world of 2050, you will need not merely to invent new ideas and products – you will above all need to reinvent yourself again and again.

For as the pace of change increases, not just the economy, but the very meaning of 'being human' is likely to mutate. Already in 1848 the *Communist Manifesto* declared that 'all that is solid melts into air'. Marx and Engels, however, were thinking mainly about social and economic structures. By 2048, physical and cognitive structures will also melt into air, or into a cloud of data bits.

In 1848 millions of people were losing their jobs on village farms, and were going to the big cities to work in factories. But upon reaching the big city, they were unlikely to change their gender or to add a sixth sense. And if they found a job in some textile factory, they could expect to remain in that profession for the rest of their working lives.